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Review

The MINDSPACE Expanded Framework (MINDSPACE X): Behavioral insights to improve adherence to psychiatric medications

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The MINDSPACE framework has made it easier to incorporate insights from behavioral science into policy, including health policy, but lacks granularity. Difficult policy problems such as adherence to psychiatric medication can benefit from judicious selection of nudges. We present a MINDSPACE Expanded Framework including 34 insights from behavioral science in the 9 MINDSPACE principles to support a more detailed integration of behavioral science into policy. We illustrate the framework with examples relating to adherence to psychiatric mediation.

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Behavioural economics is a field of study that blends insights from psychology and economics to understand how people actually make economic decisions in real-life situations. Unlike traditional economics, which assumes that individuals are fully rational and always make decisions that maximise their utility, behavioural economics recognises that humans often act irrationally due to cognitive biases (such as heuristics that simplify decision-making processes), emotions, social influences, and limited self-control [1]. Insights from behavioral economics can help people live healthier lives, but the proliferation of concepts and theories can be overwhelming for policymakers. In 2010, the MINDSPACE report from the UK Institute for Government and the cabinet office presented nine principles for policymakers: messenger factors, incentives, norms, defaults, salience, priming, affect, commitments, and ego [1–4].

Those principles are grounded in 'nudge theory' [5]. Nudges are any aspect of the choice environment that alters behaviour in a predictable way without forbidding any actions or significantly changing their economic incentives, thus steering people in a particular direction while preserving freedom of choice [5]. Evidence revealing predictable patterns (sometimes called 'biases') in human decision-making can be used to nudge citizens towards more desirable behaviour. The MINDSPACE framework brings together such psychological patterns [1]. Each principle explains specific psychological effects on behaviour which occur under certain conditions.

To ensure usability some of the richness of the behavioral economics literature was omitted in the original MINDSPACE report and over the last decade, the behavioral economics literature has expanded. MINDSPACE needs updating without abandoning the simplicity of the nine core principles. In this paper we share an Expanded Framework for MINDSPACE, which deepens each of the nine original principles, overall covering a total of 34 such effects on behavior. We illustrate this MINDSPACE Expanded Framework (MINDSPACE X) through examples of how adherence to psychiatric medication could be increased. We selected influences that have been effectively utilized in behaviour change so far, which means this list can grow as more evidence is published.

Adherence to psychiatric medication

Nonadherence to psychiatric medication is a major public policy problem and an issue of equity. Nonadherence to treatments for schizophrenia and bipolar affective disorder – affecting around two-thirds of patients – frequently leads to relapse, significant mental suffering, and detention for involuntary treatment [6]. Yet, adherence depends on the day-to-day decisions of patients. Therefore, modifying the context in which

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people make decisions, sometimes called the "choice architecture", holds promise for improving the long-term quality of life among those suffering from major mental illness [5]. In this paper, we use MINDSPACE X (see Table 1) to report how existing choice architecture interventions have been used to increase medication adherence, and to generate ideas for novel approaches to improving adherence to psychiatric medications.

Messenger X

MINDSPACE X proposes three components of messenger factors. The first (M1) relates to whether a messenger is trusted or authoritative [7]. In Cullen et al.'s *T4RP* intervention [8], a professional known to the patient and in a trusting relationship was involved in setting up adherence reminder messages, whereas other SMS-based interventions have lacked a trusted

Table 1

The MINDSPACE expanded framework.

Principles	Features
Messenger	M1. We are more likely to act on information if the messenger has authority and expertise. M2. We are more likely to act on information when the messenger is similar to ourselves. M3. Our behaviour is affected by the feelings we have for the messenger. This is due partly to the *halo effect* which occurs when one trait of a person (such as being nice) is used to inform an overall judgment of that person (such as also being smart).
Incentives	I1. We dislike losses more than we like gains of an equivalent amount. This is known as *loss aversion* and may lead to the *fear of missing out*, the *endowment effect* (we value more something that we own) and the *scarcity effect* (we value more something that is rare). I2. How we behave is influenced by how much we value an incentive, and the value of an incentive depends on where we see it from – its *reference point*. I3. How we behave is influenced by how likely we believe an outcome is; and we tend to overweight small probabilities and overestimate the probability of unlikely but easy to imagine, or easy to recall, events. I4. We behave in ways that maximise immediate gains at the expense of later gains, which means we live for today at the expense of tomorrow but tomorrow is not so strongly preferred to the day after (this is known as *hyperbolic discounting*). I5. We tend to allocate different benefits and costs to discrete mental accounts and hesitate to reallocate such resources to other priorities.
Norm	N1. We tend to do what those around us approve, and we avoid what others disapprove. N2. We tend to do what those around us are already doing. N3. Norms have stronger impact if the target audience is somehow related to the reference group executing the target behaviour.
Defaults	D1. We regularly accept what the default setting is and rarely opt out from the pre-set option (status quo bias). D2. We tend to repeat behaviours given recurring circumstances, as a result of habit formation.
Salience	S1. Our attention is drawn to what is more visible. S2. We are more likely to register something that is novel. S3. We are more likely to register information that is simple and easy to understand. S4. We are more likely to focus on things that are presented in ways that relate directly to our personal experiences. S5. Providing conspicuous personal feedback on our behaviour change can encourage that behaviour. S6. Alternatives are often compared only on a single salient attribute. S7. We prefer actions for which the probability of an outcome is known (which is known as *ambiguity aversion*).
Priming	P1. Situational cues activate memories (concepts, ideas) associated with a goal, which in turn provoke actions to achieve the primed goal, without the need for conscious intentions or awareness. P2. We tend to rely heavily on the first piece of information offered (the 'anchor'), making only a small adjustment when coming up with our answer, paying little attention to other relevant information (*anchoring effect*). P3. Asking people about what they intend to do can influence their behaviour.

Table 1 (*continued*)

Principles	Features
Affect	A1. Positive emotion motivates associated behaviours. A2. Negative emotions demotivate actions mentally associated with such feelings. A3. Negative emotions motivate actions to avoid such feelings. A4. We tend to underestimate the influence of emotions on our future behaviour (known as the *hot-cold empathy gap*).
Commitment	C1. Setting a clear goal creates psychological obligation to perform a behaviour. C2. Action plans motivate and guide a person toward performing a behaviour. C3. Making public promises to perform a behaviour increases the perceived costs for failure and hence also the likelihood of the commitment being fulfilled. C4. Our action that has an impact on an individual provokes a response that has similar effect on us, because of desire for fairness (we tend to reciprocate a positive action with another positive action, and vice versa).
Ego	E1. We tend to behave in ways that support the impression of a positive self-image (also known as *self-serving bias* and *fundamental attribution error*). E2. We try to act in ways that are consistent with the images we have of ourselves, and when beliefs and behaviour come into conflict, it is often our beliefs that get altered. E3. The greater the expectation placed on us the better we perform (which is known as the *Pygmalion effect*). E4. We tend to favourably compare ourselves against others (known as the *superiority bias*). E5. We assign more blame to someone who creates a negative outcome by doing than not doing (known as the *commission bias*).

messenger. The second messenger factor (M2) is similarity between the messenger and the audience [7]. While this approach has not been studied in a mental health context, evidence from adherence to cardiovascular system medication [9] suggests that race-congruence improves engagement among African American patients; a similar approach could be applied to psychiatric prescribing. The third messenger factor (M3) relies on the Halo Effect: we expect somebody with one positive attribute to have other unrelated positive attributes [7]. For example, kindness is expected to correlate with wisdom and somebody beautiful is expected to be providing better treatment. We are not aware of any studies which have used an intervention based on the Halo Effect to increase medication adherence.

Incentive X

Incentives have been used in the context of psychosis, depression, and substance use disorders [10**] [11]. MINDSPACE highlights the importance of incentives, but MINDSPACE X proposes 5 key behavioral insights for effective incentives drawing on Prospect Theory [12]. First, people avoid losses more than they pursue gains [13] (I1) and, similarly, the value of an incentive depends on the reference point (I2) [14]. For example, in one RCT where patients were offered incentives for take their regular antipsychotic injections non-adherence reduced by around 35 % in the first three months, but after nine months they reduced non-adherence by over 50 % [15]. This suggests that the effect of incentives increases over time, perhaps because once people expect an incentive for each antipsychotic dose, the missed incentive from an omitted dose feels like a financial loss. Loss aversion was explicitly leveraged in a pilot where participants received $15 per week but $2 was subtracted for each missed dose [16].

Prospect Theory also predicts that people overvalue small probabilities and therefore lotteries offer a more cost-effective tool (I3) [17]. A systematic review of 30 studies of attendance for psychiatric treatment found no difference between lotteries and guaranteed incentives [10**]. However, lotteries have not been utilized widely in psychiatric medication adherence and may offer an additional behavioral component.

Our tendency to overvalue immediate pleasure or pain relative to delayed pain or pleasure is one reason to use financial incentives to immediately reward behaviors which may not pay-off for weeks or months (I4) [18]. For this reason, effective incentives tend to be given either immediately or within a few days [19,20,21**]. Another reason it is important to associate the incentive with the behavior comes from mental accounting [22], the idea that people keep money for different purposes in different mental boxes (I5). While a $10 incentive may appear small in the grand scheme of things, people may compartmentalize it in a separate account to spend on a treat. This allows a small incentive to have an outsized impact on behavior, as illustrated by evidence that

financial incentives improve antipsychotic engagement [23].

Norms X

The human tendency to follow norms is reported in MINDSPACE. We identify two types of norms in MINDSPACE X: people do what others say is morally right (N1) and they do what they believe other people do (N2). The more similar a group is to us, the more powerful these norms are (N3). These insights can be leveraged to support psychiatric medication adherence: people with psychosis endorse the idea that taking antipsychotic medication is the right thing to do but the effect of this perception on adherence has not been tested [24]. There is evidence from HIV medicine that people miss more doses if they believe others have low adherence [25*]. A pilot study of adolescents with epilepsy and low medication adherence utilized this theory by testing a digital intervention allowing participants to compare their adherence against the adherence of other adolescents with epilepsy and found a small positive effect compared with an active control arm [26]. Although no such intervention has been reported in the context of adherence to psychiatric medication, a similar intervention has the potential to be effective.

Defaults X

As noted in the MINDSPACE report, people rarely opt-out of the default option (D1). Default interventions powerfully promote passive programs (such as retirement savings and organ donation) but defaults are difficult to envisage when patients must actively perform a behaviour such as medication acceptance. This difficulty is illustrated by one study which tested the effect of automatically refilling prescriptions, finding increased utilization of pharmacy services but no effect on health outcomes [27,28].

For repeated active behaviors such as medication routines, the insight that we repeat behaviour patterns and form habits is more important (D2). The link between habit and adherence is well-established in contexts such as respiratory medicine and HIV medicine ([29,30]). However intervening to promote habit formation is difficult: one study found that people took morning medications more reliably than evening medications, which may be useful for SSRIs which can be taken at different times of day but not for sedating psychiatric medications [31] Financial incentives can also be used to induce habit, but in the context of antipsychotic adherence there was no evidence of habituation lasting beyond removal of the incentive [15].

Salience X

MINDSPACE described the salience principle: "our attention is drawn to what is novel and seems relevant to us" [2]. MINDSPACE X proposes seven elements of salience and the first three relate to attention: our attention is drawn to things that are more visible (S1), we register things that are more novel (S2), and we register easy to understand information (S3). Ringtone reminders have been used to draw patients' attention to their antipsychotic treatment capitalizing well on salience because the reminder is prominent (S1), novel (S2) and unambiguous (S3) [32].

Personalization is also key: we focus on information that is personalized regarding our characteristics (S4) and our past actions (S5). Digital interventions are well-placed to provide personalized prompts so Ben-Deev et al. used data from an antipsychotic drug level-monitoring patch to provide prompts driven by patient recent adherence (S5) [33].

People make comparisons based on salient factors too; we tend to compare by a single salient factor (S6) and attempt to avoid ambiguity (S7). For example, people appear to compare daily oral antipsychotics with long-acting injectable preparations based on the salient certain discomfort of an injection, even though they recognize that monthly injectable treatment can be more effective at reducing the probability of psychotic relapse [34,35].

Priming X

Priming interventions can change behaviour through subtle olfactory cues such as hygienic smells or visual cues such as a pair of watching eyes (P1) [2]. While we could imagine a watching eyes intervention in the context of psychiatric medication adherence, interventions like these may be inappropriate to implement in a private residence. However people place excess weight on the first information they receive (P2) — sometimes called 'anchoring' [36] — and medication adherence could be promoted by focusing on the benefits of medication and risk of relapse, rather than focusing on adverse effects. Finally, people tend to follow through on intentions once they have made a plan, for example to get to the voting booth, so people may be more likely to adhere to medication if prompted to comment on where it will fit into their daily routine or how they plan to collect their prescriptions (P3) [37]. However, these priming measures have as yet been underutilized in the context of medication adherence.

Affect X

Emotional associations can drive behaviour in several ways. Positive associations motivate behaviour (A1) and negative associations de-motivate behaviour (A2) as well as driving avoidant behaviour (A3) [2]. In the context of HIV medicine, a smile, a handshake, and small cash incentive were noted to have a large affective impact, motivating adherence [38]. A similar design in mental health could encourage people to take antidepressants

and take pride in their desire to improve their mental health. Mental health staff already report seeking to associate positive affective experiences with antipsychotic injections, for example by taking patients to a café afterwards [39]*.

The cold-hot empathy gap explains that people in an emotionally 'cool' state struggle to envisage challenges they might face once in an emotionally 'hot' state (A4) [40]. It is difficult to design interventions around this but given emotional lability is inherent to the course of psychiatric illnesses, plans made in one emotional state will frequently be actioned in another emotional state. In one sense, this insight explains why people who have consented to treatment struggle to adhere amid stressful daily lives and explains the advantages of why long-acting injectable medications [41].

Commitment X

Systematic review evidence suggests that behavioral contracts can promote patient behavior change, in keeping with the idea that people prefer to act in ways consistent with prior commitments (C1) [42]. Contracts have been used to increase adherence to antihypertensives [43]. In fact, in England and Wales, formal contracts are often drawn up within the framework of Community Treatment Orders, whereby patients risk involuntary detention if they fail to adhere to treatment but these are imposed on patients not entered into voluntarily [44]. Another insight relating to commitment is that people also try to maintain two-way commitments with others (C2) [2]. In one study, people with alcohol use disorder entered a commitment to take antabuse therapy as long as their partner did not bring up past drinking behavior [45]. Such informal commitments may have the potential to increase adherence in circumstances where using legal force is inappropriate.

Ego X

MINDSPACE proposed that people try to act in ways that make them feel good about themselves. When we are praised for something, we try to continue (E1) and when we have started doing something we try to remain consistent (E2) [46]. When people have high expectations of us we try to live up to them (E3) [46]. In the context of HIV medicine, an intervention sought to promote personal pride and confidence by increasing antiretroviral drug adherence among Tanzanian adults living with HIV [47]. While ego might seem an unlikely candidate for increasing psychiatric medication engagement, interviews and surveys indicate that people prescribed antipsychotics view adherence as "good behavior" which says something positive about them and merits acknowledgement from others [24].

We tend to compare ourselves favorably against other people (E4) and we assign more blame to people whose actions cause negative outcomes compared with those whose inaction has adverse consequences (E5) [46]. This is easy to see in the context of medication non-adherence which can be viewed as more benign than, for example, amphetamine abuse, even though each can drive psychotic relapse. Using this insight to increase adherence may be difficult but, like the cold-hot empathy gap, perhaps explains why non-adherence is difficult to overcome.

Conclusions and implications

Compared with MINDSPACE, MINDSPACE X offers a richer range of distinct insights for policymakers to draw upon. In this paper we have shown that all approaches from the MINDSPACE framework have been used to increase medication adherence, most for psychiatric medications, but MINDSPACE X offers a range of as yet underutilized opportunities for increasing adherence to psychiatric medications and reducing the relapse of mental illness, and we hope this expanded framework will enable scholars in a range of other contexts to also consider how to capitalize on insights from behavioral science.

Several outstanding challenges remain for scholars and practitioners seeking to capitalize on the MINDSPACE X framework to improve engagement with psychiatric treatments. Adding nudges may be ineffective where there is excessive paperwork burden, or 'sludge', deterring and distracting people from making the healthy choice [48,49]. For example, where paperwork demands make it difficult to register with a primary care physician, patients may miss out on the opportunity for early identification and treatment of mental illness in the community [50]. Therefore, effective implementation of MINDSPACE X may also require a review of any systemic barriers to access, sometimes called a 'sludge audit' [48,51]. Likewise, there have been calls for behavioral scientists to consider the need to integrate insights from implementation science into their methods [52,53]. For example, the use of financial incentives in psychiatry has been impacted by staff feeling uncomfortable handling cash [54]. In the future, scholars utilizing MINDSPACE X should draw on the systematic approaches to implementation offered by implementation science. While the components of the MINDSPACE X framework are individually well evidenced, changing behavior in mental health services will require multidisciplinary scholarship to address outstanding sludge and integrate implementation science.

Author contribution

Conceptualization: NH, IV; Data curation: na; Formal analysis; na; Funding acquisition: IV; Investigation: NH; Methodology: IV; Project administration: IV; Resources: na; Software: na; Supervision: IV; Validation: IV, GK;

Visualization: NH; Roles/Writing – original draft: NH; Writing – review & editing: GK.

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Declaration of competing interest

NH declares that he has no interests to declare.
GK declares that he has no interests to declare.
IV declares that he has no interests to declare.

Data availability

No data was used for the research described in the article.

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* of special interest
** of outstanding interest

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Further information on references of particular interest

10. ** This systematic review included 19 studies of the how financial incentives impacted COVID-19 vaccine uptake. None found a negative effect and those conducted using high quality methods such as synthetic control were more likely to find a positive result. Incentives ranged from $5 to $5 million.

21. ** This 10-week single arm study aimed to incentivize patients to take antipsychotics, antidepressants, and mood stabilizers. 25 patients could receive $15 twice each week but lost $2 each time they missed a medication check-in. In this small pilot there was no change from pre-intervention adherence rates but the design nevertheless illustrates that loss framed incentives can be implemented in the context of adherence to psychiatric medications.

23. * This systematic review identified 26 papers including quantitative, qualitative and philosophical papers exploring antipsychotics in major mental illness. No empirical studies have suggested that incentives fail to change behavior in this context. The review presented a typology of criticisms of such incentives and outlined 19 issues requiring further empirical study.

25. * 195 people living with HIV in Uganda were surveyed about their ART adherence and their perception of local ART adherence rates. The study found a negative association between ART adherence and the belief that most other local people prescribed ART were non-adherent. They also found 76% believed other people prescribed ART locally were not adherent, indicating that social norms threatens to undermine ART adherence among this population.